JC2 H2 Prelim Paper 1 2024
Suggested Answers

Section A Cluster 1: Development, Economy and Environment

1. Resource 1 shows the growth in solar installed capacity In India from 2008 to June 2023 (in megawatts). Resource 2A is a satellite image of Kanmai 2 in Kamuthi, Tamil Nadu, before the construction of Kamuthi solar power plant in 2009. Resource 2B is a satellite image of Kanmai 2 in Kamuthi, Tamil Nadu, after the construction of Kamuthi solar power plant in 2020. Resource 3 is an excerpt taken from an article titled 'Adani's Kamuthi solar power plant casts shadow on livelihood of locals'.

(a) Cite data from Resource 1 to describe the growth in India's solar installed capacity from 2008 to June 2023. [3]

Award 1 mark for each description of the growth in India's solar installed capacity using data from Resource 1. Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- The graph illustrates the growth in India's solar installed capacity from 2008 to June 2023, showing a **significant upward trend over the 15-year period from 3 to 70100MW**. [1]
- In the first three years from 2008 – 2010, the solar installed capacity in India was **minimal**, starting at just 3 MW in 2008, and reaching 36 MW by 2010. [1]
- After that, the **growth began to accelerate**, with capacity increasing to 936 MW in 2011, and 5129 MW by 2015. [1]
- After 2015, there was a **significant surge and the upward trend continued all the way** with huge increments reaching 70100 MW by June 2023. [1]

(b) Suggest possible reasons to explain the growth in India's solar installed capacity shown in Resource 1. [5]

Award 1 mark for each suggested reason to explain the growth in India's solar installed capacity, to a maximum of 5 marks. Award a maximum of 1 additional mark for further development of each suggestion, where applicable.

Possible responses include:

- The **cost** of solar photovoltaic (PV) panels and related technology has significantly **decreased** over the years. [1] This reduction in costs made solar energy more competitive with traditional energy sources, leading to increased adoption. [1 additional mark]
- **Innovations in solar technology**, such as improved efficiency of PV panels and advancements in energy storage, have made solar power a more attractive and reliable energy source. [1]

- India has set ambitious targets for renewable energy as part of its **commitment to the Paris Agreement**. [1] The target of 100 GW of solar capacity by 2022 was a major driver, pushing both public and private sectors to invest in solar energy. [1 additional mark]
- The Indian government has implemented **policies like tax incentives and subsidies to encourage investment in solar energy**. [1] India's government launched several initiatives like the Jawaharlal Nehru National Solar Mission (JNNSM) in 2010, which aimed to promote solar energy and increase installed capacity. [1 additional mark]
- With favourable policies, **private companies, such as the Adani Group, have been able to invest and expand solar projects**, contributing to the overall growth. [1]
- India's **vast land area**, particularly in states like Rajasthan and Gujarat, has provided ample space for large solar farms, contributing to the rapid expansion of solar capacity. [1]

(c) Describe the differences in land use in Kamuthi between 2009 and 2020 shown in Resources 2A and 2B respectively. [4]

Award 1 mark for each description of difference in land use between 2009 and 2020, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- The primary change in land use is the **transformation** from a mixed-use agricultural and natural landscape to an industrial site dominated by solar panels. [1]

2009: Pre-Solar Plant
- The area in 2009 is characterised by a **mix of agricultural land and natural vegetation**. [1] The landscape includes patches of cultivated fields, interspersed with areas of shrubs and trees, indicating a predominantly rural and agricultural setting, with more established transport network including bus stops. [1 additional mark]

2020: Post-Solar Plant
- By 2020, a large portion of the area has been converted into a **solar power plant**. [1] The solar panels cover vast stretches of land, replacing the agricultural fields and natural vegetation that were present in 2009. [1 additional mark] The image shows a highly organised layout of solar panels, indicating **large-scale industrial use of the land**. [1 additional mark]

- There is a **notable reduction in natural vegetation and agricultural fields**. [1]

- The extent of the solar farms has **increased to the west of the bund** which was absent in 2009. [1]

(d) Explain how solar power plants can impact the environment in India. [6]

Award 1 mark for each explanation of how solar power plants can impact the environment in India, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

Land use and habitat disruption

- Large scale solar farms require extensive land, which can lead to the **clearing of vegetation and the disruption of local ecosystems**. [1]
- The construction of solar power plants can **fragment habitats**, making it difficult for wildlife to move freely, potentially leading to a loss of biodiversity. [1]
- In some regions, the land used for solar farms could have been used for agriculture, leading to a potential **conflict between energy production and food production**. [1]

Water use

- Solar panels, especially in arid regions, require periodic cleaning to maintain efficiency, which can **strain local water resources**. [1]
- Water abstraction for the construction and operation of the solar plant can lead to the **depletion of the groundwater table**. [1]
- Solar power plants use water for cooling, similar to traditional power plants, which can lead to **water consumption concerns in water-scarce areas**. [1]

Waste

- Solar panels have a lifespan of about 25-30 years, and the **disposal of old panels** can present a challenge for India. [1] Improper disposal can lead to the **release of hazardous materials like cadmium and lead** into the environment. [1 additional mark]

Resource extraction

- The production of solar panels involves the extraction of raw materials like silicon, silver, and rare earth metals. [1] **Mining** these materials can have significant **environmental impacts**, including habitat destruction, water pollution, and carbon emissions. [1 additional mark]

- The use of solar power can reduce the reliance on fossil fuels, which can lead to lesser greenhouse gas emissions. The **atmospheric temperature may not increase as rapidly as before**. [1]

(e) With reference to Resources 2A and 3, describe the ecosystem services that Kamuthi can provide prior to the construction of the solar power plant. [6]

Award 1 mark for each description of the ecosystem services that Kamuthi can provide prior to the construction of the solar power plant, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

Provisioning services

- The land in Kamuthi was likely used for agriculture, providing crops and livestock that supported **local food production**. [1] For example, the goats in Resource 3 can provide milk, meat and wool. [1 additional mark]
- Kamuthi **provided water resources** for sustaining agriculture and supporting daily life in the community. [1] Resource 3 shows that natural water bodies such as kanmais (seasonal ponds), ooranis (drinking water ponds), and streams provided water for irrigation, drinking, and other domestic uses. [1 additional mark]
- The large area shown in Resource 2A provided **pastureland for livestock grazing**, which was an important part of the local economy, particularly for herders and farmers who relied on livestock for their livelihoods. [1]

Regulating services

- The natural landscape, including wetlands, **helped regulate water flow**, storing water during the rainy season and releasing it gradually, which ensured a steady water supply during dry periods. [1] The kanmai in Resource 2A retains the monsoon water which backs up and spreads out over the lands to irrigate the catchment. [1 additional mark]
- Vegetation in the area, including trees, grasslands, and crops, contributed to local **climate regulation** by sequestering carbon dioxide, providing shade, and helping to moderate temperature and humidity levels. [1]
- Natural vegetation helped **prevent soil erosion** by stabilising the soil with root systems, maintaining soil fertility, and preventing the loss of topsoil during heavy rains or strong winds. [1]

Supporting services

- The large area provided **habitat for various species of wildlife**, including birds, insects, and small mammals, supporting biodiversity and ecological balance in the region. [1]
- The area likely **supported a variety of plant and animal species**, contributing to rich biodiversity. [1]

Cultural services (can't tell from resource)

- The natural landscape may have held **cultural, spiritual, or historical significance for the local communities**. [1] Natural heritage, spiritual sense of belonging, traditional knowledge, and associated customs are important for creating a sense of belonging. [1 additional mark]

(f) With reference to Resource 3, explain the socio-economic trade-offs associated with solar power. [6]

Award 1 mark for each explanation of the socio-economic trade-offs associated with solar power, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

<u>Economic Benefits vs. Local Livelihoods</u>

- The construction of large-scale solar power plants, like the Kamuthi plant, can bring **significant economic benefits, such as job creation, infrastructure development, and increased energy security**. [1] These projects also contribute to national goals for renewable energy, potentially reducing dependency on fossil fuels and lowering greenhouse gas emissions. [1 additional mark]

- On the flip side, the displacement of agricultural land for solar power projects can **disrupt the livelihoods of local communities**, particularly those dependent on farming and grazing. [1] In Kamuthi, the fencing off of thousands of acres reduced access to grazing grounds, which directly affected the economic sustenance of farmers and herders. [1 additional mark] Promised job opportunities may also not materialize as expected, leading to social unrest and economic insecurity among those who lost their land. [1 additional mark]

<u>Renewable Energy Goals vs. Ethical Land Acquisition</u>

- Meeting ambitious renewable energy targets often requires the **rapid acquisition of land and the fast-tracking of projects**. [1] This is essential for transitioning to a low-carbon economy and ensuring energy security. [1 additional mark]

- The Kamuthi case highlights the risks of **inadequate consultation with local communities** and the potential for **unethical land acquisition practices**. [1] Allegations of illegal land acquisition and the blocking of access to communal lands (kanmais, ooranis, streams) underscore the need for transparent and ethical practices. [1 additional mark] When local communities feel disenfranchised, it can lead to protests, legal battles, and long-term resentment, undermining the social license to operate for such projects. [1 additional mark]

Section A Cluster 2: Tropical Environments

2. Resource 4 is a photograph of the karst landscape in Tam Coc National Park, Ninh Binh, Vietnam. Resource 5 shows the climograph of Ninh Binh in Vietnam. Resource 6 is a photograph of Thien Cung Cave, Ha Long Bay, Vietnam.

(a) Describe the characteristics of the karst landform A in Tam Coc National Park, Ninh Binh, Vietnam, shown in Resource 4. [4]

Award 1 mark for each description of the characteristics of the karst landform, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each description, where applicable.

Possible responses include:

- A is a tower karst of roughly 200m or more in height [1].
- It is a steep sided mass of limestone with a serrated summit [1] – left middle and foreground.
- It rises abruptly above wide flat alluvial plains (valley floors) [1] – background
- The near vertical walls are heavily fluted and grooved with deep gullies found on the surface [1]
- It is greyish in colour probably with a high percentage of magnesium. [1]
- Vegetation is more abundant at the base of the tower karst. [1]
- The landform has undergone human modification with the building of a staircase and a pagoda at the top. [1]

(b) With reference to Resource 5, explain how the karst landform A shown in Resource 4 is formed over time. [6]

Award 1 mark for each explanation of the formation of karst landform A, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

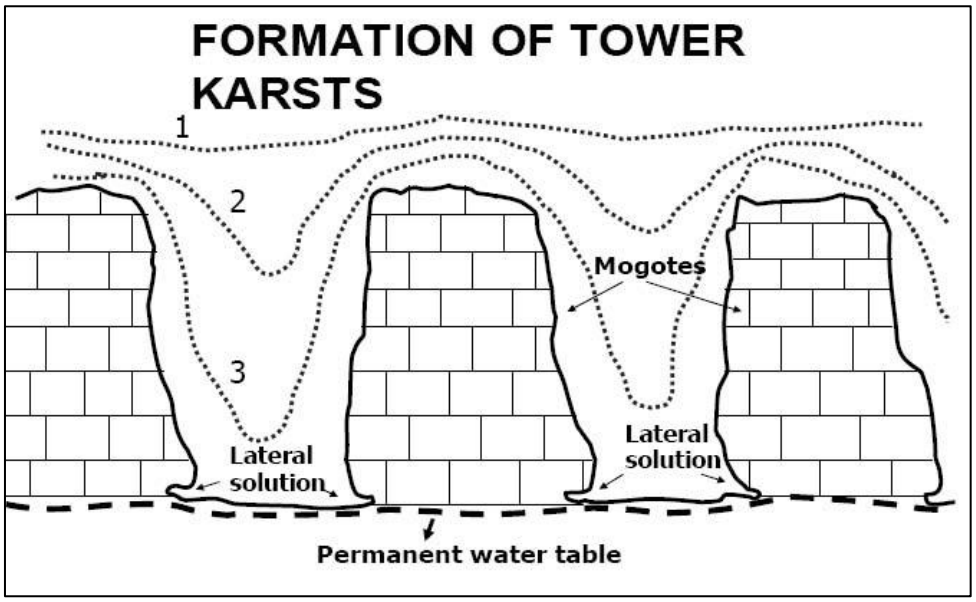
Possible responses include:

- A is probably a product of a combination of **downward and lateral erosion** [1], where **dissolution** weakens the limestone bedrock, and the **more resistant**/ durable portions of the rock **remains** (less jointed). [1 additional mark]

- The limestone is susceptible to **carbonation** especially during the cooler months of Nov to Mar (R5) when **acidulated rainwater** comes into contact with it and weathers it down [1]. During the wetter months from Jul to Sep (R5), the **calcium carbonate** which is weathered to become calcium bicarbonate, is removed in **solution**. [1 additional mark]

- There is **downward erosion** until the base level is reached. The river lost its potential energy to downcut and will start to cut sideways instead. [1] The slopes of the hills undergo continuous **parallel retreat (scarp retreat)** via the processes of

weathering, mass movements and surface wash. A flat alluvial plain seen in R4 appears until tower karst is formed. [1 additional mark] Refer to diagram (draw/ describe) [1 additional mark]



- Often, **complex features** like caves and sinkholes and subterranean rivers can be found within, forming a complex geological environment. [1]

(c) With reference to Resource 6, explain how feature B is formed. [4]

Award 1 mark for each explanation of how feature B is formed, to a maximum of 4 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

- Feature B is a column, formed when stalactites and stalagmites unite or are jointed together. [1]

- Stalactites are formed when water charged with calcium bicarbonate enters joins and evaporates as it drips of the celling of caves, leaving behind a small amount of calcium carbonate [1]. The water also drips on the ground directly below it to form stalagmites. [1 additional mark]

- This process of deposition takes place over a long period of time, and is also dependent on the amount of calcium carbonate that is present in the water. [1]

(d) With reference to Resources 4 and 6, explain how human activities can accelerate the weathering processes in the karst landscapes in Vietnam. [6]

Award 1 mark for each explanation of how human activities can accelerate the weathering processes in the karst landscapes in Vietnam, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

- Human activities can accelerate the weathering processes in karst landscapes in Vietnam by accelerating or modifying the processes of karst formation.

- Firstly, human activities like deforestation can reduce vegetation cover and increase surface runoff [1]. These will alter drainage patterns and affect rates of carbonation and solution, reducing the karstification processes when lesser water can enter through joints and bedding plains to carve out caves and caverns like that shown in R6. [1 additional mark]

- Secondly, when there is excessive groundwater withdrawal by humans, it destabilises underground voids and increases sinkhole formation [1]. It can also slow down chemical weathering and erosional processes on carbonate rocks that are essential for karst landform formation, like caves and caverns shown in R6. [1 additional mark]

- Thirdly, tourism can bring about interference to carbonation and erosional processes when pathways (stairs) like those in R4 are constructed on the tower karst [1]. These will either seal the entry of water over those areas of limestone or artificially direct rainwater to cause more weathering and erosional processes over certain areas of the tower karst. [1 additional mark]

- Tourism can also introduce more carbonation and erosion when more carbon dioxide is release around limestone regions by increased human or vehicular traffic [1]. This can lead to increased carbonation processes that accelerate karst landscape formation. [1 additional mark]

- Tourism however, can also have a positive impact by increasing awareness, funding conservation efforts, and supporting scientific research, fostering appreciation for karst features, prompting conservation initiatives funded by tourism revenue [1].

(e) With reference to Resource 6, suggest the ecosystem services that may be provided by the cave. [6]

Award 1 mark for each suggestion of the ecosystem services that may be provided by the cave, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each suggestion, where applicable.

Possible responses include:

- Caves like that in R6 provide vital habitats (regulating services) for a variety of specialized species, including cave-dwelling bats, invertebrates, and plants adapted to low-light conditions [1]. Pest predators live in the caves in the karst area, like insectivorous bats that can eat approximately 6,000 mosquitoes, serving as a pest predator and also playing its role in fruit pollination. [1 additional mark]

- The beauty of karst regions is also often marketed as sites of tourist attractions (cultural services) [1]. Caves like those n R6 offer unique experiences for physical, mental, and aesthetic rejuvenation, at the same time contributing to local economies and promoting cultural exchange. Visitors learn about unique geological features, traditional practices, and conservation efforts, fostering a sense of global citizenship and environmental stewardship. [1 additional mark]

- Karst landscapes like those in R6 also provide various supporting ecosystem services that contribute to biodiversity conservation [1]. Temperature gradients offer niches for diverse species, humid pockets shelter moisture-loving life, and filtered sunlight nurtures a rich botanical tapestry. [1 additional mark]

(f) Explain how human activities can affect ecosystem services provided by karst landforms. [6]

Award 1 mark for each explanation of how human activities can affect the ecosystem services provided by karst landforms, to a maximum of 6 marks. Award a maximum of 1 additional mark for further development of each explanation, where applicable.

Possible responses include:

- Human activities like mining and quarrying can destroy karst landscapes and affect the ecosystem services provided by karst landforms [1]. These activities can leave dep scars on the karst landscape that often penetrate the bedrock, damaging the intricate network of subterranean channels that transport water through the aquifer. [1 additional mark]

- Human activities like deforestation can also affect the ecosystem services provided by karst landforms when vegetation are removed on a large scale, driven by urbanisation or agriculture [1]. The reduced vegetation cover leads to increased surface runoff, bypassing vital aquifer, affecting the water provisioning services. **OR** Eroded soil in karst regions can clog up sinkholes and cave systems, impacting water quality and disrupting sensitive cave ecosystems. **OR** Sedimentation can smother cave formations and threaten the survival of cave-dwelling organisms, further jeopardizing the biodiversity and ecological services provided by karst landscapes (any one). [1 additional mark]

- Unsustainable tourism practices like unregulated cave exploration and inappropriate lighting can disrupt delicate cave ecosystems, harming cave-dwelling organisms like bats and invertebrates. [1] Improper waste disposal from tourists can introduce pollutants into cave ecosystems, impacting water quality and harming cave-dwelling organisms. [1 additional mark]

Cluster 1 Development, Economy and Environment

3. Evaluate the success of strategies to manage impacts of extractive industries on places. [20]

Possible Approaches:

Candidates could approach the question by making a judgement on whether some strategies are **more successful** than others through a **consideration** of the relative strengths and limitations of the strategies. The strategies could be aimed at addressing **different impacts** such as **economic, environmental and social.** The strategies could be compared in relation to **criteria** such as extent of benefits, and ability to address root causes of the impacts. Candidates could also **analyse** the challenges in managing impacts of extractive industries on places.

Candidates could also approach the question by making a judgement on whether the **same strategies** have been **more successful in some places** than others through a consideration of **two or more case studies**. Candidates could **evaluate** whether the strategies in each case achieved the intended aims. Candidates could also **analyse** the **contextual factors** such as the scale of the impacts, governance, degree of institutional quality and the resources available, which may have contributed to the variations in success between places.

Levels marked using Generic Level Descriptors for 20m H2 essays

4. 'The Global Production Networks (GPNs) of Transnational Corporations (TNCs) positively impact host economies more than home economies.' Evaluate this statement. [20]

Possible Approaches:

Candidates could approach the question by making a judgment on whether the Global Production Networks (GPNs) of Transnational Corporations (TNCs) positively impact host economies more than home economies through a consideration of criteria such as **spatial scale of the impacts, the temporal scale of the impacts and the stakeholders impacted**. Candidates should consider a range of **economic, social and environmental impacts**, **both positive and negative**, to evaluate the significance of impacts.

Candidates could approach the question by making a judgement on whether the GPNs of TNCs positively impact host economies more than home economies **in some context than others** through a consideration of two or more case studies. Candidates could analyse the impacts in terms of **contextual factors such as the development level of the economies, power relations between the state and TNCs, nature of TNC activities** etc.

Levels marked using Generic Level Descriptors for 20m H2 essays

Cluster 2 Tropical Environments

5. 'Synoptic scale circulation contributes to the distinctive rainfall patterns in the tropics.' Evaluate this statement. [20]

Possible Approaches:

Candidates could approach the question by making a judgment on whether synoptic scale circulation (influence of distribution of heat over land and sea on summer and winter monsoon circulations, tropical cyclones, El Niño Southern Oscillation) contributes to the distinctive rainfall patterns in the tropics through a consideration of criteria such as the **scale** of that contribution in terms of **space and time** and the **resultant effect** on the climate of the region. Candidates could also consider the contribution of global scale circulation (atmospheric circulation in the Hadley cells, seasonal migration of the Hadley cells) and evaluate the **contribution of both** synoptic and global scale circulation to the distinctive rainfall patterns in the tropics.

Candidates could also approach the question by making a judgement on whether synoptic scale circulation contributes to the distinctive rainfall patterns in the tropics **in some contexts than others** through a consideration of two or more case studies. Candidates could analyse the **contextual factors** such as latitude, location (e.g. coastal vs inland, cyclone pathway), physical characteristics (e.g topography), etc. which may have contributed to the differences in the relative influence of the circulations.

Levels marked using Generic Level Descriptors for 20m H2 essays

6. 'Fluvial deposition has the greatest impact on fluvial landforms in the humid tropics.' Evaluate this statement. [20]

Possible Approaches:

Candidates could approach the question by making a judgment on whether fluvial deposition has the greatest impact on fluvial landforms in the humid tropics through a consideration of **criteria such as the type and scale** of the fluvial landforms (eg. meanders, braided streams, deltas). Candidates should also consider other fluvial and geomorphic processes and analyse **which process has the greatest impact** on fluvial landforms.

Candidates could approach the question by making a judgment on whether fluvial deposition has the greatest impact on fluvial landforms **in some context** (eg. meanders vs deltas) than others through a consideration of two or more case studies. Candidates could **evaluate the relative significance of other fluvial and geomorphic processes** and analyse the **interplay** between the processes.

Levels marked using Generic Level Descriptors for 20m H2 essays